TRIASHA SARKAR

Atlanta, GA · +1 (404) 948-8761 · tsarkar34@gatech.edu
linkedin.com/in/triasha-sarkar · github.com/triasha72 · triasha72.github.io/Portfolio

Machine learning engineer and Georgia Tech aerospace MS graduate who builds evaluation-first systems from data and experiments through deployment. My recent work connects retrieval and ranking, LLM evaluation, distributed training, high-throughput serving, and policy-conditioned moderation.

EDUCATION

MS, Aerospace Engineering | Georgia Institute of Technology, Atlanta, GA **Aug 2026**

Graduate Research Assistant, Aerospace Systems Design Laboratory (ASDL)

MSc, Aerospace Vehicle Design | Cranfield University, United Kingdom **Feb 2023**

First Class Honours · Thesis on geometry optimization under FAA constraints

B.Tech, Aerospace Engineering | SRM Institute of Science and Technology, India **Jun 2021**

First Class Honours

TECHNICAL SKILLS

ML & GenAI: PyTorch, FSDP, DeepSpeed, Megatron-LM, Transformers, vLLM, SGLang, TensorRT-LLM, RAG, hybrid retrieval, reranking, PEFT/LoRA, GRPO evaluation

Systems: Python, Go, SQL, Kafka, FastAPI, Docker, Kubernetes, GitHub Actions, PostgreSQL/pgvector, Prometheus, OpenTelemetry, ONNX, Core ML, Qualcomm QNN

Scientific computing: NumPy, SciPy, pandas, NetworkX, OSMnx, statistical inference, design of experiments, surrogate modeling

PROFESSIONAL EXPERIENCE

Graduate Research Assistant under Prof. Dimitri Mavris | Georgia Tech ASDL **May 2025 – Aug 2026**

- Took simulation, safety, and sustainability questions from data review and model design through tested analysis, with reproducible evidence for faculty and sponsors.
- Checked assumptions with domain experts, documented limitations, and delivered models and workflows that others could rerun and use.

Machine Learning Engineer | Rolls-Royce India **Jul 2023 – Apr 2025**

- Built Python workflows for diagnostics, anomaly detection, predictive maintenance, and mixed-frequency time series from multivariate engine data.
- Turned certification requirements into traceable analyses and model checks, reviewing each result with lifecycle engineers before use.

Data Science Intern | Rolls-Royce DataLabs **May – Jul 2021**

- Cleaned aircraft-engine sensor data, designed features, and compared predictive and anomaly-detection models to identify recurring failure patterns.

SELECTED TECHNICAL PROJECTS

AeroRAG-X | Retrieval, Post-Training, and ML Systems **2025 – Present**

github.com/triasha72/AeroRAG-X

- Built hybrid BM25/dense retrieval, RRF, cross-encoder reranking, pgvector search, evidence gating, and citation controls over 3,233 source-preserving NASA NTRS chunks.
- Added protected Base/LoRA evaluation, bounded agents, FastAPI services, observability, FSDP/DeepSpeed/Megatron training treatments, and vLLM/SGLang/TensorRT-LLM serving paths. GPU result tables remain pending.

IntegrityBench | Policy-Conditioned Moderation Evaluation **Aug 2026 – Present**

github.com/triasha72/IntegrityBench

- Built a 360-case benchmark across policy shifts, long context, multilingual, adversarial, standard, and low-resource slices with structured allow/reject/escalate decisions.
- A static-policy control reached 0.620 macro F1 and 0.286 false acceptance; current-policy retrieval reached 1.000 on the controlled generator. I report this as a construction check, not production moderation evidence.

NewsLens | Recommendation, Real-Time Search, and Production ML **Jul – Aug 2026**

github.com/triasha72/NewsLens

- Used chronological splits, TF-IDF ranking, and a training-only fallback for leakage-safe MIND recommendation, reaching NDCG@10 of 0.366; the one-time holdout interval included zero.
- Built a Go, Kafka, PostgreSQL, and FastAPI article path with idempotency and dead letters. A 500-event local run measured 44 ms publish p99, 79 ms freshness p95, and 5.7 s partition recovery.

EdgeGenBench | Scientific ML and On-Device Inference **Aug 2026 – Present**

github.com/triasha72/EdgeGenBench

- Built an aircraft-design surrogate benchmark with uncertainty estimates, constrained optimization, physics checks, ONNX export, and deployment tests.
- The Qualcomm QNN model retained 0.997 mean held-out R^2; a drifting INT8 candidate was rejected by policy. Shipped an installable Safari app without claiming native-device performance.

AeroSynth-Eval | Multimodal AI Evaluation **Aug 2026 – Present**

github.com/triasha72/AeroSynth-Eval

- Built a fixed 48-scenario VLM suite with deterministic assets, provenance checks, annotation workflows, and typed evaluator contracts.
- A Kaggle P100 batch attempted all 12 development cases: eight passed the response schema and four JSON failures were retained for diagnosis. The run proves execution behavior, not human alignment.

Silent Failure Detection in Rocket-Motor Simulations **May – Jul 2026**

Georgia Tech ASDL

- Audited a design sweep that failed without errors, built a leakage-safe classifier with 96.8% accuracy across 15,120 simulations, and traced the failures to an uncapped convergence loop.
- Tested unseen geometries in a separate sampling study and wrote a patch that converts silent hangs into logged convergence errors.

HERO | Source-Aware Retrieval for Airline Safety **2025 – 2026**

Georgia Tech ASDL

- Worked on retrieval and evaluation methods that keep generated safety answers tied to the technical sources analysts need to review.

GREEN TEA and Project EAGLE | Sustainable Aviation Modeling **Sep 2025 – Apr 2026**

Georgia Tech ASDL

- Built surrogate, uncertainty, demand, and life-cycle models for alternative-fuel and transport studies; the GREEN TEA model remains in sponsor use.

Atlanta Mobility Resilience Digital Twin **2026 – Present**

github.com/triasha72/atlanta-mobility-resilience-digital-twin

- Built an OSMnx and NetworkX simulator for road-closure effects on Atlanta travel time and access, with deterministic core tests in CI and live OSM download kept as an explicit integration step.

Equity Backtest | Walk-Forward Signal Evaluation **Jul 2026**

github.com/triasha72/Equity-Backtest

- Built an expanding-window backtest with transaction costs and retained every specification tested; the apparent momentum edge disappeared after costs and missed the preregistered significance threshold.

LEADERSHIP

Graduate Chair | Women of Aeronautics and Astronautics **2025 – 2026**

Georgia Tech